

Replication of Study 2 by Jamie Amemiya, Gail D. Heyman, and Caren Walker (2024, Cognitive Science)

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Introduction

Study 2 of Amemiya et al. (2024) finds that adults who wanted to persuade others to support or reject a public policy selectively presented comparison cases that encouraged the counterfactual inference consistent with their position. Specifically, participants in the *Motivated to Support* condition were more likely to show the supporting comparison cases than participants in the *Motivated to Reject* condition, and that participants in the *Motivated to Reject* condition

were more likely to show the rejecting comparison cases than participants in the *Motivated to Support* condition.

I chose this study because I’m interested in the development of our ability to understand that the way evidence was selected affects how much weight we should give it. Past work has shown that young children are sensitive to how samples are generated (Kushnir et al., 2010) and can strategically select true evidence to deceive others (Rhodes et al., 2015). These studies, along with the target study, primarily approach the problem from the perspective of the person producing or selecting evidence, but I aim to also investigate the less-studied child-as-consumer side and to examine the relation between these two sides of the problem: 1) understanding how to select evidence to persuade someone, and 2) recognizing that the process by which true evidence is selected affects the value of that evidence (how informative it is). This project would provide an important foundation for my future work by establishing whether the evidence-selection behavior observed in adults can be replicated before extending the paradigm to investigate how children might produce and evaluate selectively presented evidence.

Participants will read a vignette about a novel virus spreading in five regions: Hawaii (US), Hokkaido (Japan), Te Waipounamu (New Zealand), Oregon (US), and California (US). In this vignette, leaders were unsure how to respond, so they designed a set of policies (“Policy A”). Only Hawaii implemented Policy A, on Day 5. The other four regions waited to see whether it worked. Participants will be randomly assigned to imagine being a politician who either strongly believes Policy A would be effective (*Motivated to Support* condition), or strongly believes Policy A would be ineffective (*Motivated to Reject* condition). Participants will then view a full five-location graph and describe the data in their own words. The graph is a figure that plots new virus cases over Days 0-10 for all five regions, with a vertical line at Day 5 marking the intervention. Hawaii’s cases are shown to drop sharply right after Day 5, so there was strong temporal covariation between policy and decline in every condition. Oregon and California kept climbing after Day 5, implying the supporting counterfactual (Hawaii’s cases would have kept rising without the policy). Hokkaido and Te Waipounamu declined on their own after Day 5, implying the rejecting counterfactual (Hawaii’s cases would have fallen anyway).

Participants will then choose one of the four following graphs to show on the nightly news to support their assigned position: (a) Target only: Hawaii’s trajectory alone (b) Target + Supporting Comparisons: Hawaii plus Oregon and California (c) Target + Rejecting Comparisons: Hawaii plus Hokkaido and Te Waipounamu (d) Target + All Comparisons: all five regions They will also provide an open-ended response justifying their decision.

This study will be implemented using Qualtrics. There will also be several attention/comprehension and manipulation checks throughout the survey. Anyone who fails a comprehension check will be excluded. One potential challenge I anticipate encountering is exclusions for failing comprehension checks. The procedure is a bit demanding (there are two training modules, a news story with attention checks and feedback, and several open-ended responses), so I am a

little concerned I may lose a decent number of participants if students rush through the study, since it is an unpaid/course credit survey.

See the project repository [here](#) and the original paper [here](#).

Methods

Power Analysis

Original effect size, power analysis for samples to achieve 80%, 90%, 95% power to detect that effect size. Considerations of feasibility for selecting planned sample size.

Planned Sample

Planned sample size and/or termination rule, sampling frame, known demographics if any, preselection rules if any.

Materials

All materials - can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Procedure

Can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Analysis Plan

Can also quote directly, though it is less often spelled out effectively for an analysis strategy section. The key is to report an analysis strategy that is as close to the original - data cleaning rules, data exclusion rules, covariates, etc. - as possible.

Clarify key analysis of interest here You can also pre-specify additional analyses you plan to do.

Differences from Original Study

Explicitly describe known differences in sample, setting, procedure, and analysis plan from original study. The goal, of course, is to minimize those differences, but differences will inevitably occur. Also, note whether such differences are anticipated to make a difference based on claims in the original article or subsequent published research on the conditions for obtaining the effect.

Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or “none”.

Results

Data preparation

Data preparation following the analysis plan.

Confirmatory analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Replication Attempt

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.